Solution Of One-Dimensional Heat Equation

We know that one dimensional heat equation is

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2} \quad \text{where} \quad c^2 = \frac{k}{s\rho} \quad \text{---(1)}$$

(s is the specific heat of the material, ρ is the density and k is the thermal conductivity)

The above heat equation can be written as

$$u_{xx} - \frac{1}{c^2} u_t = 0$$

Here $A = 1, \; B = 0, \; C = 0$

$$B^2 - 4AC = 0 - 0 = 0$$

so the equation is Parabolic.

Let us solve, by the method of finite differences

Consider a rectangular mesh in the x-t plane with spacing h along x-direction and k along time t direction. Denoting the mesh point $(x, t) = (ih, jk)$ as simply i, j we have

$$\frac{\partial u}{\partial t} = \frac{u_{i,j+1} - u_{i,j}}{k}$$

$$\frac{\partial^2 u}{\partial x^2} = \frac{u_{i-1,j} - 2u_{i,j} + u_{i+1,j}}{h^2}$$

$$(1) \Rightarrow \frac{u_{i,j+1} - u_{i,j}}{k} = \frac{c^2}{h^2}\left[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right]$$

$$\Rightarrow u_{i,j+1} - u_{i,j} = \frac{kc^2}{h^2}\left[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right]$$

$$\Rightarrow u_{i,j+1} - u_{i,j} = \alpha\left[u_{i-1,j} - 2u_{i,j} + u_{i+1,j}\right]$$

$$\boxed{u_{i,j+1} = \alpha\left[u_{i-1,j} + u_{i+1,j}\right] + (1-2\alpha)u_{i,j}} \quad \text{---(2)}$$

where $\alpha = \frac{kc^2}{h^2}$ is the mesh ratio parameter

This formula enables us to determine the value of u at the $(i, j+1)^{th}$ mesh point in terms of the known function values at the x_{i-1}, x_i and x_{i+1} at the instant t_j.

It is a relation between the function values at the two levels $j+1$ and j, therefore called a 2-level formula.

The above formula is called Schmidt explicit formula which is valid for $0 < \alpha \leq 1/2$

Note :- When $\alpha = 1/2$ in (2)

$$\boxed{u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right]}$$

and the above formula is called Bendre-Schmidt formula.

(2) Find the values of $u(x,t)$ satisfying the Parabolic equation $\frac{\partial u}{\partial t} = 4\frac{\partial^2 u}{\partial x^2}$ and the boundary Conditions $u(0,t) = 0 = u(8,t)$ and $u(x,0) = 4x - \frac{x^2}{2}$ at the points $x = i$, $i = 0,1,2,\ldots 8$ and $t = \frac{j}{8}$, $j = 0,1,2,\ldots 5$

Solutions:- Here $c^2 = 4$, $h = 1$, $k = 1/8$

then $\alpha = \frac{c^2 k}{h^2} = \frac{1}{2}$

∴ We have Bendre - Schmidt's method

$$u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right] \quad \text{——(1)}$$

Since $u(0,t) = 0 = u(8,t)$

$u_{0,j} = 0$ and $u_{8,j} = 0$ for all values of j

i.e, entries in the first and last column are zero.

$$u(x,0) = 4x - \frac{x^2}{2}$$

$$u_{i,0} = 4i - \frac{i^2}{2}$$

$u_{0,0} = 0$, $u_{3,0} = 7.5$, $u_{6,0} = 6$,

$u_{1,0} = 3.5$, $u_{4,0} = 8$, $u_{7,0} = 3.5$

$u_{2,0} = 6$, $u_{5,0} = 7.5$,

put $j = 0$ in (1)

$$u_{i,1} = \frac{1}{2}\left[u_{i-1,0} + u_{i+1,0}\right]$$

For $i = 1, 2, \ldots 7$ we get

$u_{1,1} = \frac{1}{2}[u_{0,0} + u_{2,0}] = \frac{1}{2}[0+6] = 3$

$u_{2,1} = \frac{1}{2}[u_{1,0} + u_{3,0}] = 5.5$

$u_{3,1} = \frac{1}{2}[u_{2,0} + u_{4,0}] = 7$

$u_{4,1} = \frac{1}{2}[u_{3,0} + u_{5,0}] = 7.5$

$u_{5,1} = \frac{1}{2}[u_{4,0} + u_{6,0}] = 7$

$u_{6,1} = \frac{1}{2}[u_{5,0} + u_{7,0}] = 5.5$

$u_{7,1} = \frac{1}{2}[u_{6,0} + u_{8,0}] = 3$

put $j=1$ in (1)

$$u_{i,2} = \frac{1}{2}[u_{i-1,1} + u_{i+1,1}]$$

For $i = 1, 2, \ldots 7$ we get

$u_{1,2} = \frac{1}{2}[u_{0,1} + u_{2,1}] = \frac{1}{2}[0+5.5] = 2.75$

$u_{2,2} = 5$

$u_{3,2} = 6.5$

$u_{4,2} = 7$

$u_{5,2} = 6.5$

$u_{6,2} = 5$

$u_{7,2} = 2.75$

$u_{8,2} = 0$

i / j	0	1	2	3	4	5	6	7	8
0	0	3.5	6	7.5	8	7.5	6	3.5	0
1	0	3	5.5	7	7.5	7	5.5	3	0
2	0	2.75	5	6.5	7	6.5	5	2.75	0
3	0	2.5	4.625	6	6.5	6	4.625	2.5	0
4	0	2.3125	4.25	5.5625	6	5.5625	4.25	2.3125	0
5	0	2.125	3.9375	5.125	5.5625	5.125	3.9325	2.125	0

(2) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ subject to the conditions $u(x,0) = \sin \pi x$; $0 \le x \le 1$

$u(0,t) = u(1,t) = 0$. Carry out computations for two levels taking $h = 1/3$ and $k = 1/36$

Solution :- Here $c^2 = 1$, $h = 1/3$, $k = 1/36$

$$\alpha = \frac{kc^2}{h^2} = \frac{1}{4}$$

$u(x,0) = \sin \pi x$ $\quad u(x_i, 0) = \sin \pi x_i = \sin \pi(ih) = \sin\left(\frac{i\pi}{3}\right)$

$u_{1,0} = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$

$u_{2,0} = \sin \frac{2\pi}{3} = \sqrt{3}/2$

$$u_{3,0} = \sin\left(\frac{3\pi}{3}\right) = \sin\pi = 0$$

$$u(0,t) = 0 \Rightarrow u_{0,1} = 0 \qquad u_{0,2} = 0$$

$$u(1,t) = 0 \Rightarrow u_{1,1} = 0 \qquad u_{1,2} = 0$$

Schmidt's Formula

$$u_{i,j+1} = \alpha u_{i-1,j} + (1-2\alpha) u_{i,j} + \alpha u_{i+1,j}$$

$$= \frac{1}{4}\left[u_{i-1,j} + 2u_{i,j} + u_{i+1,j}\right]$$

j \ i	0	1	2	3
0	0	$\sqrt{3}/2$	$\sqrt{3}/2$	0
1	0	0.6495	0.6495	0
2	0	0.4871	0.4871	0

$i=1,\ j=0$

$$u_{1,1} = \frac{1}{4}\left[u_{0,0} + 2u_{1,0} + u_{2,0}\right] = \frac{1}{4}\left[0 + 2\times\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2}\right]$$

$$= 0.6495$$

$i=2,\ j=0$

$$u_{2,1} = \frac{1}{4}\left[u_{1,0} + 2u_{2,0} + u_{3,0}\right] = \frac{1}{4}\left[\frac{\sqrt{3}}{2} + 2\times\frac{\sqrt{3}}{2} + 0\right]$$

$$= 0.6495$$

$i=1,\ j=1$

$$u_{1,2} = \frac{1}{4}\left[u_{0,1} + 2u_{1,1} + u_{2,1}\right]$$

$$= \frac{1}{4}\left[0 + 2(0.6495) + 0.6495\right] = 0.4871$$

$i=2,\ j=1$

$$u_{2,2} = \frac{1}{4}\left[u_{1,1} + 2u_{2,1} + u_{3,1}\right]$$

$$= \frac{1}{4}\left[0.6495 + 2(0.6495) + 0\right] = 0.4871$$

(3) use schmidt method to solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ under the conditions

(i) $u(0,t) = u(1,t) = 0, \quad t \geq 0$

$u(x,0) = \sin \pi x, \quad 0 < x < 1$

taking $h = 1/4$ and $\alpha = 1/6$. Carry out three steps in the time level.

Solution :- Since $h = 1/4$, the given interval $[0,1]$ on which x-varies, is to be divided into 4 equal parts, with 1 to 4.

$x_i = ih$

$x_i = \frac{i}{4}$

$x_1 = 1/4$

$x_2 = 2/4$

$x_3 = 3/4$

$x_4 = 1$

$u(0,t) = u(1,t) = 0, \ t \geq 0$ gives

$u(0,t) = 0 \Rightarrow u_{0,j} = 0 \quad j = 0,1,2$

$u(x_4,t) = 0 \Rightarrow u_{4,j} = 0 \quad j = 0,1,2$

Given initial Condition $u(x,0) = \sin \pi x$, $0 < x < 1$

$$u(x_i, 0) = u_{i,0} = \sin \pi x_i = \sin \pi \left(\frac{i}{4}\right)$$

$$= \sin\left(i \frac{\pi}{4}\right)$$

$$u_{1,0} = \sin \pi/4 = 1/\sqrt{2}$$

$$u_{2,0} = \sin \frac{\pi}{2} = 1$$

$$u_{3,0} = \sin \frac{3\pi}{4} = 1/\sqrt{2}$$

We note that, here $c = 1$, $h = 1/4$ and $\alpha = 1/6$

$$\therefore \quad \alpha = \frac{kc^2}{h^2} \Rightarrow k = \frac{1}{96}$$

We required to compute the solution upto 3rd level in time i.e, upto $t = 3k = 1/32$

$$u_{i,j+1} = \alpha \left[u_{i-1,j} + u_{i+1,j}\right] + (1-2\alpha) u_{i,j}$$

$$= \frac{1}{6}\left[u_{i-1,j} + u_{i+1,j}\right] + \left(1 - \frac{1}{3}\right) u_{i,j}$$

$$u_{i,j+1} = \frac{1}{6}\left[u_{i-1,j} + u_{i+1,j}\right] + \frac{2}{3} u_{i,j}$$

$$= \frac{1}{6}\left[u_{i-1,j} + u_{i+1,j} + \frac{2}{3} u_{i,j}\right]$$

j \ i	0	1	2	3	4
0	0	$1/\sqrt{2}$	1	$1/\sqrt{2}$	0
1	0	0.63807	0.90236	0.63807	0
2	0	0.57578	0.81427	0.57578	0
3	0	0.51957	0.73478	0.51957	0

$$u_{1,1} = \frac{1}{6}\left[0 + 4 \times \frac{1}{\sqrt{2}} + 1\right] = 0.63807$$

$$u_{2,1} = \frac{1}{6}\left[\frac{1}{\sqrt{2}} + 4 \times 1 + \frac{1}{\sqrt{2}}\right] = 0.90236$$

$$u_{3,1} = \frac{1}{6}\left[1 + 4 \times \frac{1}{\sqrt{2}} + 0\right] = 0.63807$$

$$u_{1,2} = \frac{1}{6}\left[0 + 4 \times (0.63807) + 0.90236\right] = 0.57578$$

$$u_{2,2} = \frac{1}{6}\left[0.63807 + 4\,(0.90236) + 0.63807\right] = 0.81427$$

$$u_{3,2} = \frac{1}{6}\left[0.90236 + 4\,(0.63807) + 0\right] = 0.57578$$

$$u_{1,3} = \frac{1}{6}\left[0 + 4\,(0.57578) + 0.81427\right] = 0.51957$$

$$u_{2,3} = \frac{1}{6}\left[0.57578 + 4\,(0.81427) + 0.57578\right] = 0.73478$$

$$u_{3,3} = \frac{1}{6}\left[0.81427 + 4\,(0.57578) + 0\right] = 0.51957$$

(4) Consider the heat equation $\frac{\partial^2 u}{\partial x^2} = \frac{\partial u}{\partial t}$ under the following conditions

(i) $u(0,t) = u(4,t) = 0$; $t \geq 0$

(ii) $u(x,0) = x(4-x)$; $0 < x < 4$

Employ the Bendre - Schmidt method with $h = 1$ to find the solution of the equation for $0 < t \leq 1$

Solution :- Initial Conditions

$u(0,t) = 0 \Rightarrow u(0,0) = u(0,1) = u(0,2) = u(0,3) = u(0,4) = 0$

$u(4,t) = 0 \Rightarrow u(4,0) = u(4,1) = u(4,2) = u(4,3) = u(4,4) = 0$

$u(x,0) = x(4-x) \Rightarrow u(1,0) = 1(4-1) = 3$

$u(2,0) = 2(4-2) = 4$

$u(3,0) = 3(4-3) = 3$

Here $c^2 = 2$, $\alpha = 1/2$, $h = 1$

$\alpha = \frac{kc^2}{h^2} \Rightarrow \frac{1}{2} = \frac{k(2)}{1} \Rightarrow k = 1/4$

$u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right]$

i / j	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	3/2	2	3/2	0
3	0	1	3/2	1	0
4	0	3/4	1	3/4	0

5 Find the numerical solution of the heat equation $\frac{\partial^2 u}{\partial x^2} = 2\frac{\partial u}{\partial t}$ when $u(0,t) = 0 = u(4,t)$ and $u(x,0) = x(4-x)$ by taking $h=1$. Find the values upto $t=5$.

Solution:- $u_t = \frac{1}{2} u_{xx}$

$c^2 = \frac{1}{2}$, $h=1$, $k = \frac{h^2}{2c^2} = 1$

The values of x in $0 \le x \le 4$ with $h=1$

Consider the initial conditions

$u(x,0) = x(4-x)$

$u_{1,0} = 3$; $u_{2,0} = 4$; $u_{3,0} = 3$

$u(0,t) = 0 \Rightarrow u_{0,0} = u_{0,1} = u_{0,2} = u_{0,3} = u_{0,4} = u_{0,5} = 0$

$u(4,t) = 0 \Rightarrow u_{4,0} = u_{4,1} = u_{4,2} = u_{4,3} = u_{4,4} = u_{4,5} = 0$

Consider $u_{i,j+1} = \frac{1}{2}\left[u_{i-1,j} + u_{i+1,j}\right]$

when $j=0$ $u_{i,1} = \frac{1}{2}\left[u_{i-1,0} + u_{i+1,0}\right]$

$i=1$, $u_{1,1} = \frac{1}{2}\left[u_{0,0} + u_{2,0}\right] = \frac{1}{2}\left[0+4\right] = 2$

$i=2$, $u_{2,1} = \frac{1}{2}\left[u_{1,0} + u_{3,0}\right] = \frac{1}{2}\left[3+3\right] = 3$

$i=3$, $u_{3,1} = \frac{1}{2}\left[u_{2,0} + u_{4,0}\right] = \frac{1}{2}\left[4+0\right] = 2$

j \ i	0	1	2	3	4
0	0	3	4	3	0
1	0	2	3	2	0
2	0	1.5	2	1.5	0
3	0	1	1.5	1	0
4	0	0.75	1	0.75	0
5	0	0.5	0.75	0.5	0